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PMC11020166 | 光氧化还原催化 | 采集状态 imported_unchecked

标题	Decarboxylative Nickel- and Photoredox-Catalyzed Aminocarbonylation of (Hetero)Aryl Bromides
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出处	Organic Letters 2023
DOI	10.1021/acs.orglett.3c02389
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC11020166/
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摘要（原文节选）

An efficient methodology for the photoredox- and nickel-catalyzed aminocarbonylation of (hetero)aryl bromides was developed. The utilization of readily available oxamic acids, the application of a broadly used organic photoredox catalyst (4CzIPN), and mild reaction conditions make this transformation an appealing alternative to classical amidation procedures. The generation of carbamoyl radicals was supported by trapping reactions with a hydrogen atom transfer catalyst in the presence of D2O, yielding the deuterate ⋯

数据表 · Table 1（21 行，含数值行 21 行）

Optimization of the Reaction Conditions

Entrya	PC	PC (mol %)	Base	Solvent	Concentration	Modifications	Conversionb	Yieldc (Isolated yield)
1	Cl-TXO	20	Li2CO3	DMF	33 mM		39%	25%
2	Cl-TXO	20	Na2CO3	DMF	33 mM		49%	22%
3	Cl-TXO	20	K2CO3	DMF	33 mM		58%	28%
4	Cl-TXO	20	Cs2CO3	DMF	33 mM		49%	23%
5	Cl-TXO	20	Li2CO3	DMF	33 mM	Phthalimided	56%	27%
6	di-tBu-Mes-Acr+ BF4 –	5	Li2CO3	DMF	33 mM	SynLED	0%	0%
7	(Ir(dF(CF3)ppy) 2(dtbpy))PF6	8	Li2CO3	DMF	33 mM		76%	31%
8	4CzIPN	1	Li2CO3	DMF	33 mM		78%	53%
9	4CzIPN	3	Li2CO3	DMF	33 mM		>95%	63% (51%)
10	4CzIPN	5	Li2CO3	DMF	33 mM		>95%	45%
11	4CzIPN	8	Li2CO3	DMF	33 mM		>95%	47%
12	4CzIPN	15	Li2CO3	DMF	33 mM		>95%	40%
13	3CzClIPN	3	Li2CO3	DMF	33 mM		0%	0%
14	5CzBN	3	Li2CO3	DMF	33 mM		>95%	63%
15	3DPA2FBN	3	Li2CO3	DMF	33 mM		44%	23%
16	4CzIPN	3	Li2CO3	DMF	33 mM	No H2O	>95%	45%
17	4CzIPN	3	Li2CO3	THF	33 mM		0%	0%
18	4CzIPN	3	Li2CO3	DMC	33 mM		0%	0%
19	4CzIPN	3	Li2CO3	CH3CN	33 mM		0%	0%
20	4CzIPN	3	Li2CO3	DMF	20 mM		>95%	36%
21	4CzIPN	3	Li2CO3	DMF	50 mM		>95%	64%

脚注（逐字）：aReaction conditions: 4-Bromobenzonitrile (18.2 mg, 100 μ mol), 1a (35.8 mg, 200 μ mol), base (200 μ mol), NiCl2 · glyme (2.20 mg, 10.0 μ mol), 4,4 ′ -dtbbpy (4,4-di-tert-butyl-2,2-dipyridyl) (3.22 mg, 12.0 μ mol), argon, 22 h, Kessil tuna blue as a light source (λ max = 464 nm).bConversion based on 1H NMR an

正文对应句（逐字）：The amide bond is an exceptionally important structural motif not only as the backbone of peptides but also as the linkage of numerous small bioactive pharmaceuticals and agrochemicals. Owing to the high abundance of amides in the final bioactive products1,2 or synthetic intermediates, amidation rea

数据表 · Table 2（10 行，含数值行 10 行）

Deuterated Formamide Synthesis: Optimizationa

Entryb	PC	Thiol (mol %)	D2O (equiv)	Yieldc (D incorporation)
1	[Ir]	10	28	37% (73%)
2	[Ir]	10	56	40% (87%)
3	4CzIPN	10	28	22% (43%)
4	4CzIPN	10	56	43% (90%)
5	4CzIPN	10	112	11% (92%)
6	4CzIPN	30	56	35% (87%)

Entryb	PC	Thiol (mol %)	D2O (equiv)	Yieldc (D incorporation)
7	4CzIPN	60	56	32% (86%)
8	–	10	56	0%
9	4CzIPN	10	–	44% (0%)
10	4CzIPN	–	56	0%

脚注（逐字）：a[Ir] = Ir(dF(CF3)ppy)2(dtbpy))PF6.bReaction conditions: 1a (17.9 mg, 100 μ mol), 4CzIPN (2.37 mg, 3.00 μ mol), Li2CO3 (7.39 mg, 100 μ mol), DMF (2.0 mL), argon, 22 h, Kessil tuna blue (λ max = 464 nm).cYields were determined by the 1H NMR analysis of the crude mixture using 1,2,4,5-tetramethylbenzene a

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